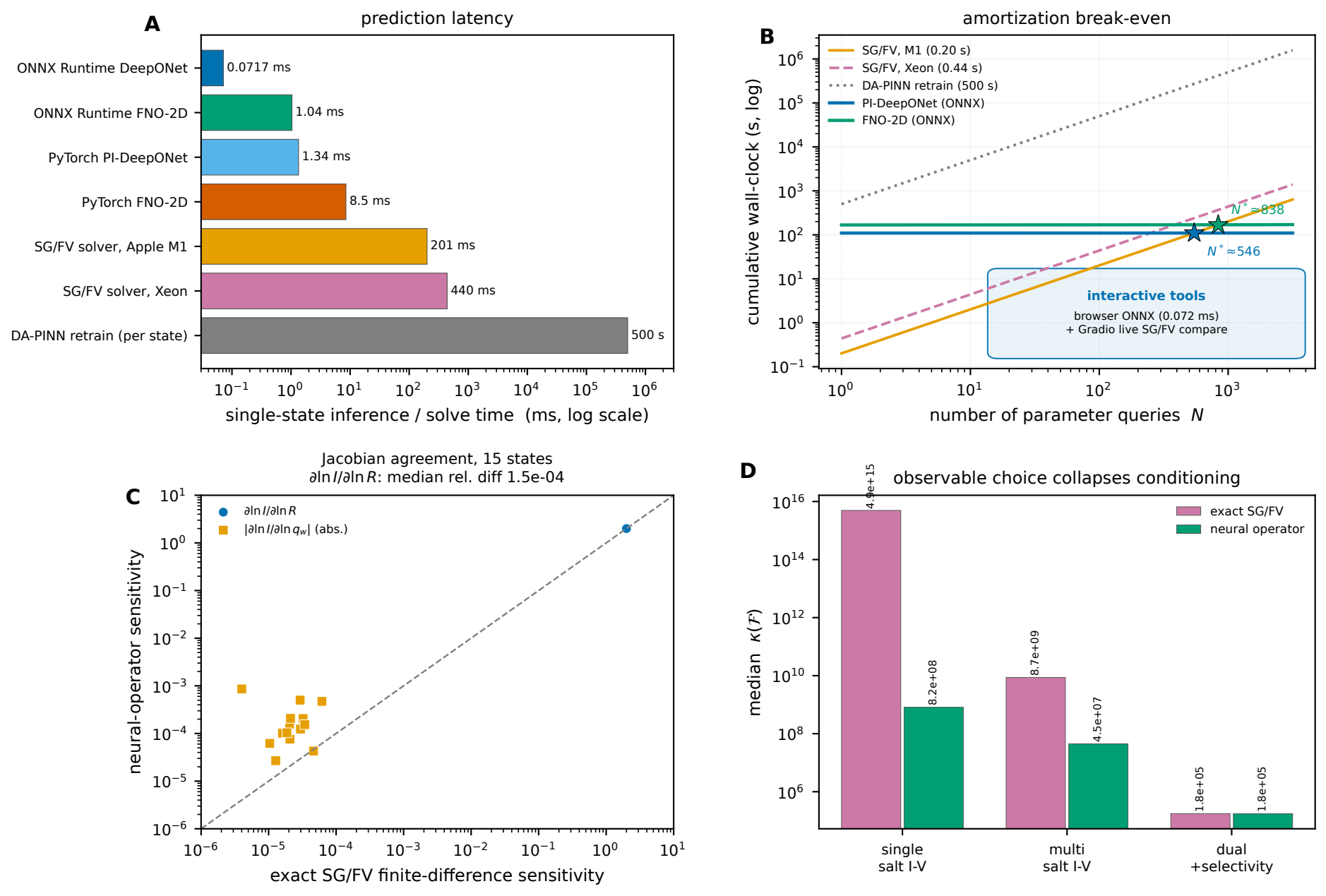
A
prediction latency
ONNX Runtime DeepONet
ONNX Runtime FNO-2D
PyTorch PI-DeepONet
PyTorch FNO-2D
SG/FV solver, Apple M1
SG/FV solver, Xeon
DA-PINN retrain (per state)
0.0717 ms
1.04 ms
1.34 ms
8.5 ms
201 ms
440 ms
500 s
single-state inference / solve time (ms, log scale)
B
amortization break-even
SG/FV, M1 (0.20 s)
SG/FV, Xeon (0.44 s)
DA-PINN retrain (500 s)
PI-DeepONet (ONNX)
FNO-2D (ONNX)
N* ≈838
N* ≈546
interactive tools
browser ONNX (0.072 ms)
+ Gradio live SG/FV compare
cumulative wall-clock (s, log)
number of parameter queries N
C
Jacobian agreement, 15 states
∂ln I/∂ln R: median rel. diff 1.5e-04
∂ln I/∂ln R
|∂ln I/∂ln q_w| (abs.)
neural-operator sensitivity
exact SG/FV finite-difference sensitivity
D
observable choice collapses conditioning
exact SG/FV
neural operator
4.9e+15
8.2e+08
8.7e+09
4.5e+07
1.8e+05
1.8e+05
median κ(F)
single
salt I-V
multi
salt I-V
dual
+selectivity